

Estimand Divergence Between Static Docking and Molecular Dynamics as a Triage Filter for Antimalarial Leads

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Abstract

Using a curated collection derived from 396 African natural products and 454 synthetic antimalarials, we evaluated a multi-target docking framework across 136 wild-type and mutant receptor states of PfDHFR and PfCRT. On a primary candidate cohort ($n = 12$), a Resistance-Resilience Score (RRS) provided a five-tier stratification decoupling predicted wild-type binding strength from mutant-state score retention. Explicit-solvent molecular dynamics (MD) simulations showed a directional divergence in 87.5 % (7/8; 95% Wilson CI 52.9–97.8 %, $n = 8$) of matched candidate–target states, wherein dynamic pocket relaxation retained poses where static grid docking predicted

score penalties. At single-replicate scope this divergence is a protocol-local calibration signal: static docking scores and short MD trajectories provide complementary, non-equivalent triage metrics for natural product lead optimization.

Introduction

Drug resistance remains the principal threat to malaria control in Africa. An estimated 282 million cases and 610 000 deaths occurred globally in 2024, with the WHO African Region bearing the greatest burden¹. Partial resistance to artemisinin derivatives, mediated by mutations in *Pfkelch13*, is now confirmed or suspected in at least eight African countries², and declining efficacy of some partner drugs has been reported^{1,3}. Spatial-temporal modelling further indicates that the geographic footprint of *pfk13* markers in Africa is expanding⁴, while genomic surveillance documents the co-occurrence of *Pfcrtr* and *Pfdhfr* resistance variants across diverse transmission settings⁵⁻⁷.

African natural products are an underexplored antimalarial chemical space: rigid polycyclic frameworks, macrocyclic systems, and rare functional groups challenge scoring functions calibrated on synthetic drug-like molecules⁸. Under a single-drug-multi-target framing, simultaneous retention of binding across multiple key parasite targets is hypothesized to raise the mutational barrier to treatment failure⁹, and natural products are attractive starting points because their stereochemical complexity supports activity across distinct protein classes^{9,10}. However, static docking grid scores capture only rigid pocket complementarity and often fail to account for side-chain reorganization or solvent-mediated accommodation upon mutation^{8,11}.

The upstream library used for this study was generated with generative molecular methods and dual-filter consensus docking (AutoDock Vina¹² and DiffDock-L¹³), yielding 65 856 candidate molecules from 396 African natural products and 454 synthetic drugs (upstream library record; Supporting Information Table S16). We selected the candidate panel against a receptor set spanning established resistance determinants and mechanistically distinct para-

site processes: PfDHFR (dihydrofolate reductase, PDB 7F3Y) and PfCRT (chloroquine resistance transporter, PDB 6UKJ) represent clinically established resistance-associated contexts; PfATP4 (sodium efflux pump, PDB 9N10) probes ion homeostasis; and PfClpR (PDB 4GM2) represents a proteostasis-related target. This panel provides a mechanistic cross-section to evaluate whether resistance-aware prioritization remains informative across distinct target classes.

Specifically, this study addresses a central research question: *Does a multi-target, resistance-aware computational workflow (docking-derived RRS + PNS + ACSI) provide a discriminant triage of African natural products across drug-resistant mutant panels—and how does short explicit-solvent molecular dynamics structurally re-calibrate static docking predictions?*

To answer this question, we present three methodological components: (i) reporting *Estimand Divergence* between static docking RRS and explicit-solvent MD pose retention as a protocol-local triage observation, wherein dynamic pocket accommodation was consistent with retained poses in 87.5 % (7/8; 95% Wilson CI 52.9–97.8 %, $n = 8$) of comparable mutant states; (ii) applying the pre-declared WT-eligibility rule that excludes non-binding wild-type baselines ($|S_{WT}| < 5.0 \text{ kcal mol}^{-1}$) consistently to prevent artificial score inflation; and (iii) testing protocol robustness across an independent external sensitivity panel of 39 compounds (312 docking complexes) rescored by GNINA convolutional neural network (CNN) affinity scoring with 100 % class-level agreement (no per-mutant rank transfer claimed). Rather than viewing the directional mismatch between docking and short MD as a computational flaw, we report short explicit-solvent MD as a secondary structural check that flags static grid artifacts for prospective biochemical testing; at single-replicate scope this mismatch is a calibration signal, not a validated prediction.

Methods

Study design

The primary analysis is docking-score retention (RRS) for the 12 Set-C candidates that have eligible PfDHFR and PfCRT wild-type scores. RRS is the operational baseline against which the central comparison—the directional divergence between docking-RRS and short-MD structural retention—is tested. We therefore distinguish the primary docking-RRS analysis from the central divergence comparison, secondary structural-retention diagnostics, technical validation, and unobserved biological quantities. The 16-system Set-C MD pilot provides the matched-row comparison for the divergence signal; the five PfCRT-only candidates and the four-complex parent-study MD check are reported separately as secondary analyses. They address different coverage and sampling questions and are not combined with the primary RRS classes. The parent-study MD systems are not among the 17 Set-C candidates and do not validate the Set-C ranking.

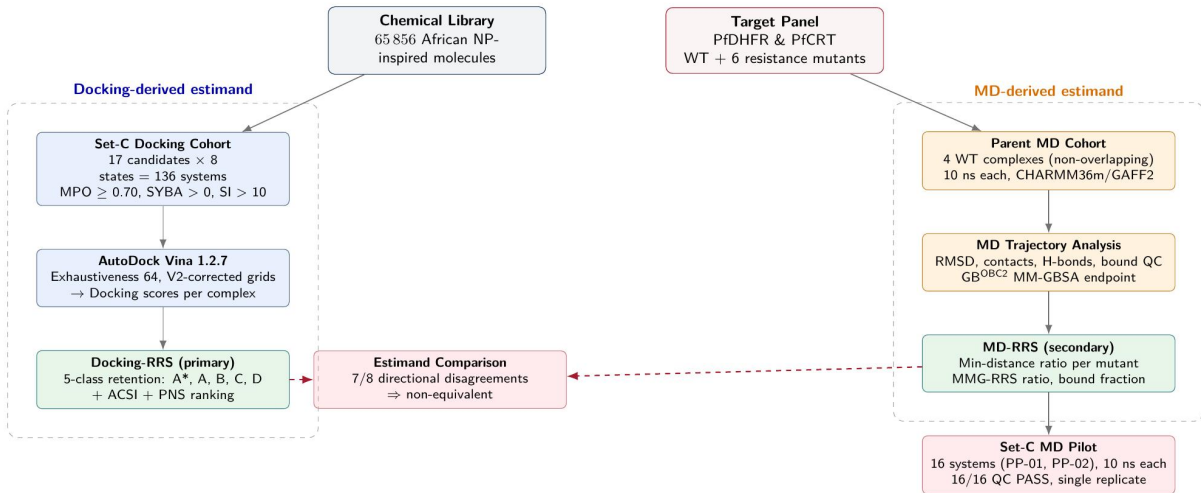


Figure 1: Integrated Resistance-Aware Polypharmacology and Molecular Dynamics Triage Framework. Static multi-target docking (PfDHFR, PfCRT, PfATP4, PfClpP) with Resistance-Resilience Scores (RRS), Antimalarial Chemical Space Index (ACSI), and Polypharmacological Network Score (PNS), followed by 10 ns explicit-solvent MD (OpenFF 2.2.0/CHARMM36m) as a secondary check on the protocol-local Estimand Divergence signal (7/8 matched states). **PP-01** passes the two-target pilot gate; **PP-02** remains single-target scope and was not promoted; **PP-15** is a wild-type feasibility probe outside the Set-C estimand with no RRS estimate. Resistance or evolutionary interpretations require prospective experimental validation.

Library provenance and source-class counts follow the upstream library record (Supporting Information Table S16). Primary candidates were prioritized using multi-parameter optimization ($\text{MPO} \geq 0.70$, retaining 19 913 primary leads), predicted synthetic accessibility ($\text{SYBA} > 0$), predicted selectivity index ($\text{SI} > 10$), target coverage across at least two docking receptors, and scaffold diversity (≤ 3 compounds per Bemis–Murcko scaffold). This selection yielded the Set-C candidate panel ($n = 17$), which includes one previously named consensus lead (Ligand 87, PfDHFR). Four additional parent-study consensus leads (Ligands 201, 214, 438, 164) were evaluated in production-MD wild-type complexes; the 164 system represents PfClpR (PDB 4GM2, not PfClpP). Five approved antimalarial drugs (artemisinin, pyrimethamine, chloroquine, lumefantrine, cipargamin) served as reference controls (Table 1). Class frequencies are conditional on these filters and should not be extrapolated to the full library.

Table 1: Breakdown of the 136 docking protein–ligand systems by target. The 17 Set-C candidates were docked against 5 PfDHFR states (WT + N51I, C59R, S108N, I164L) and 3 PfCRT states (WT + K76T, K76A).

Category	Ligands	Targets	Systems	MD (ns)
PfDHFR WT + 4 mutants (Set-C)	17	5	85	—
PfCRT WT + 2 mutants (Set-C)	17	3	51	—
Docking total	—	—	136	—

The 136 docking systems were not subjected to production MD. The 16-system pilot and four non-overlapping parent-study complexes were analyzed separately.

Resistance mutation panel

We selected six mutant or comparator states (Table 2); target identifiers are in the Supporting Information (Table S14). PfDHFR substitutions and PfCRT K76T represent resistance-associated contexts described in WHO resources; K76A is a mechanistic comparator, not a prevalence estimate.

Table 2: *Resistance mutation panel.*

Target	State	Resistance context	Evidence boundary
PfDHFR	N51I	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	C59R	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	S108N	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	I164L	High-level antifolate-resistance state	Comparator for severe resistance
PfCRT	K76T	Chloroquine-resistance marker	Transporter-resistance context
PfCRT	K76A	Mechanistic comparator for Lys76 perturbation	No clinical prevalence or resistance-marker claim

Homology modelling of resistance mutants

No mutant crystal structures were available; homology models were generated with SWISS-MODEL from wild-type templates PfDHFR (7F3Y) and PfCRT (6UKJ). PfATP4 and the 164-labelled parent-MD system were not part of the panel; PDB 4GM2 is PfClpR, not PfClpP. Models were retained only when QMEAN exceeded -4.0 , GMQE exceeded 0.7 , Ramachandran favored residues exceeded 95% , and backbone RMSD was below $1.5\text{ \AA}$. All six models met these thresholds (Table S2). Model scores are not experimental structures.

Docking protocol assessment

The docking protocol follows the upstream library workflow (Supporting Information Table S16): AutoDock Vina 1.2.7¹² with exhaustiveness 64 against grids centered on the corrected V2 binding-site coordinates¹⁴ of PfDHFR (7F3Y), PfCRT (6UKJ¹⁵), PfATP4 (9N10), and the historically labelled PfClpR/4GM2 entry (not PfClpP). The V2 grid specification represents the post-review refinement from the companion study, including the critical

PfCRT K76→LYS restoration for wild-type reference state and P2Rank-validated binding-site centers. P2Rank¹⁶ pocket predictions were compared with the grid centers as a boundary audit of these fixed boxes (Figure S1).

Parent-study enrichment benchmarks (V2 grids) are summarized here and tabulated in full in the Supporting Information (Table S1). On the MMV Malaria Box consensus (Vina+DiffDock), ROC-AUC was 0.924 (PfDHFR), 0.971 (PfCRT), and 1.000 (PfATP4). In contrast, DEKOIS 2.0 enrichment for PfDHFR with Vina alone was null (ROC-AUC 0.450, $EF_{5\%} = 0.00$), indicating that score-based recovery of known actives is protocol- and library-dependent. Redocking was considered accurate when heavy-atom RMSD was $< 2.0 \text{ \AA}$ relative to the crystal pose; the MTX redock attempt was a clear failure ($\text{RMSD} \approx 30 \text{ \AA}$). AutoDock Vina was used for all reported Set-C scores; GNINA yielded no usable score for this panel. Docking scores are empirical scoring approximations that are sensitive to the scoring function and protocol^{8,17,18}; they should not be treated as binding free energies.

Secondary docking calibration

A secondary Tartarus cross-docking analysis used QuickVina on the primary-lead panel. The composite MPO–docking-score association was null and exploratory; complete correlations are provided in the Supporting Information (Tables S9 and S10).

African Natural Product chemical space and structural descriptors

The candidate panel (Set-C, $n = 17$) was derived from the African Natural Products Database (ANPDB) and synthetic antimalarial reference sets. Physicochemical and stereochemical descriptors were computed using RDKit (v2025.03.6). The cohort exhibits a mean molecular weight (MW) of $(271.9 \pm 69.3) \text{ g/mol}$, a partition coefficient (LogP) of 2.84 ± 0.82 , a quantitative estimate of drug-likeness (QED) of 0.70 ± 0.06 , a fraction of sp^3 -hybridized carbons (F_{sp^3}) of 0.22 ± 0.15 , and an average of 0.6 ± 0.9 chiral centers per scaffold.

These properties reflect compact, low-molecular-weight natural product cores (includ-

ing prenylated chromones, flavonoid derivatives, and indole alkaloids) isolated from African medicinal flora such as *Cryptolepis sanguinolenta*, *Enantia chlorantha*, and *Artemisia* species. The high mean MPO score (0.728 ± 0.021) and synthetic accessibility scores (SYBA = 36.6 ± 23.4) indicate chemical-space alignment with synthetic feasibility.

MPO sensitivity and predicted ADMET profile

MPO-weight sensitivity and single-source ADMET-AI predictions were used descriptively; the complete sensitivity and endpoint results are provided in the Supporting Information (Figure S2 and Table S3). These predictions were not experimentally validated.

Docking-derived resistance-retention score definition

All 17 candidates were docked against each wild-type and mutant structure using AutoDock Vina 1.2.7 with exhaustiveness 64. The 136 systems were ranked with empirical Vina scores rather than calibrated binding free energies. Grid boxes were centered on the co-crystallized ligand position with 25 Å dimensions, matching the audited V2 configuration files¹⁴.

For target t , let $s_{\text{Vina},i,\text{WT},t}$ and $s_{\text{Vina},i,m,t}$ denote the signed Vina scores. Candidates with $|s_{\text{Vina},i,\text{WT},t}| < 5.0 \text{ kcal mol}^{-1}$ are excluded from that target’s score-retention analysis. The target-specific resistance-retention score (RRS) is

$$\text{RRS}_{i,m,t} = \frac{|s_{\text{Vina},i,m,t}|}{|s_{\text{Vina},i,\text{WT},t}|} \times 100. \quad (1)$$

RRS is a relative docking-score-retention statistic, not a free-energy difference or resistance measurement. The available-target summary averages the defined target-specific values across eligible targets and mutant states. The primary analysis is restricted to the 12 candidates with eligible PfDHFR and PfCRT wild-type scores; the five PfCRT-only candidates are reported separately.

The mutually exclusive classes are defined from the available mutant values; primary

class counts use the complete two-target subset:

- **Class A***: all available mutant RRS values are at least 80 % and the minimum eligible wild-type score magnitude is at least 7.0 kcal mol⁻¹.
- **Class A**: all available mutant RRS values are at least 80 %, but the minimum eligible wild-type score magnitude is below 7.0 kcal mol⁻¹.
- **Class B**: all available mutant RRS values are at least 70 %, but at least one is below 80 %.
- **Class C**: at least one available mutant RRS value is at least 80 %, but not all available values reach 70 %.
- **Class D**: no available mutant RRS value reaches 80 %.

The classes are mutually exclusive. The cut-offs follow the Vina score distribution in this panel and are not calibrated to biochemical resistance or clinical outcome. Sensitivity to alternative thresholds is in the Supporting Information (Table S5).

Secondary ranking descriptors

ACSI places each candidate relative to approved-drug and African natural-product reference spaces; weight-sensitivity checks are in the Supporting Information (Table S4). PNS is a network-weighted ranking score; PfCRT centrality was imputed, with sensitivity reported in the Supporting Information (Table S8). Associations among metrics were assessed by Spearman rank correlation with 100 000 permutations, Bonferroni adjustment for three tests, and bootstrap intervals from 10 000 resamples. At $n = 12$ these correlations are descriptive and are not powered tests of independence.

MD protocol

We simulated all four parent-study wild-type systems for 10 ns with CHARMM36m¹⁹/GAFF2²⁰ at 310.15 K using GROMACS^{21,22}. The 10 ns length is intended as a pose-retention and local-geometry check, not as a measurement of residence time or converged free energy²³. We analyzed the trajectories after removing periodic-boundary discontinuities. MDAnalysis²⁴ was used to calculate contacts, H-bonds, RMSF, and radius of gyration every 10 frames and RMSD on every frame.

MM-GBSA endpoint estimates were obtained from production trajectories using the GB^{OBC2} model (igb=5) with 0.15 M salt concentration, via the `gmx_MMPBSA` v1.5.0.3²⁵ CHARMM36-to-AMBER conversion. These are endpoint enthalpic scores, not rigorous free energies, and their limitations for ranking dissimilar complexes are well documented²⁶. Predicting mutation effects via endpoint methods remains challenging, with best-case correlations around 0.44 even at 100 ns²⁷. Configurational entropy is not included. MM-GBSA endpoint values are reported with the frame-level SEM for the interpretable parent-study and PP-15 summaries, whereas the Set-C table displays the propagated within-trajectory standard deviation (SD_{prop}); neither quantity is a replicate-level uncertainty estimate. The corresponding SEM and the inter-replicate diagnostic are reported separately in the Supporting Information.

The 16-system Set-C pilot used the same temperature and length, with ligand parameters from OpenFF 2.2.0 (AM1-BCC) rather than GAFF2. Absolute MM-GBSA values are therefore not compared between the two cohorts. Each mutant was run as a single replicate; trajectories were accepted for endpoint analysis only after geometric quality checks.

Results

Operational reproducibility of the docking-RRS signal

Mean target-specific RRS was 75.1 for PfDHFR (12 candidates) and 86.2 for PfCRT (17 candidates); full target-wise summaries are in the Supporting Information (Table S6). The difference reflects docking-score retention between panels, not biological resistance.

On the complete two-target set ($n = 12$) the classes were A*: 1, A: 1, B: 4, C: 5, and D: 1. On all 17 candidates classified on eligible targets, the counts become A*: 5, A: 1, B: 5, C: 5, and D: 1; the five PfCRT-only compounds are not comparable with the two-target set. Per-compound RRS profiles are shown in Figure S3. Varying the wild-type eligibility threshold from $4.0 \text{ kcal mol}^{-1}$ to $5.0 \text{ kcal mol}^{-1}$ left class counts unchanged; raising retention thresholds from 70 %/80 % to 80 %/90 % increased Class D from 1 to 9 (Table S5).

The docking-RRS signal is reproducible under the declared protocol: PP-01 multi-seed dispersion is $\pm 0.05 \text{ kcal mol}^{-1}$ for PfDHFR and $\pm 0.04 \text{ kcal mol}^{-1}$ for PfCRT, and PP-15 multi-seed dispersion is $\pm 0.03 \text{ kcal mol}^{-1}$. In a bounded score-perturbation audit, 77.9 % of RRS classes remained unchanged under $\pm 1 \text{ kcal mol}^{-1}$ perturbation. This characterizes protocol-level consistency, not pose accuracy or binding fidelity.

Chemical-space and network descriptors (secondary)

The mean ACSI for the 17-candidate cohort was 0.543, with two candidates (11.8 %) exceeding 0.70 on the approved-drug/African-NP reference space; weight-sensitivity checks are in the Supporting Information (Table S4). PNS was only weakly associated with RRS on the two-target set ($\rho = -0.2098$, adjusted $p = 1.0000$); PfCRT centrality was imputed, with sensitivity reported in the Supporting Information (Table S8). At $n = 12$ the wide bootstrap intervals indicate that the three ranking dimensions capture distinct information rather than redundant signals.

Cross-metric associations (secondary, post-selection)

These secondary, post-selection comparisons test whether docking-derived RRS aligns with PNS and ACSI within the same cohort; they are not independent validation of any mechanism.

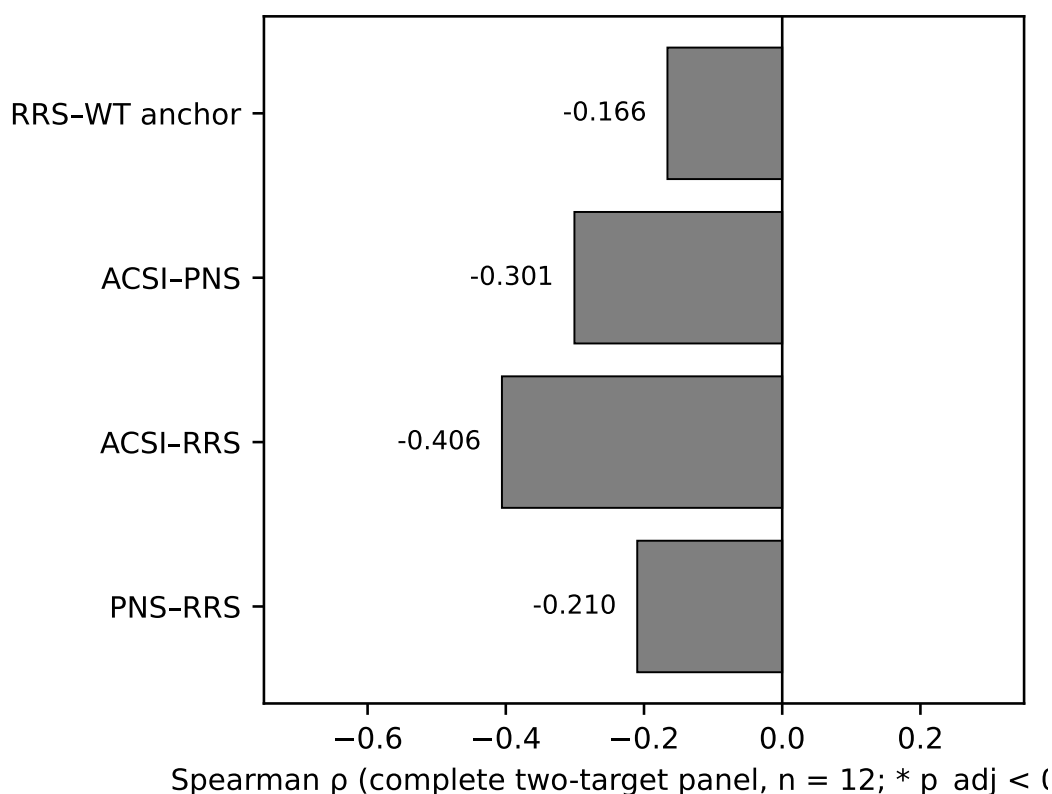


Figure 2: Cross-metric Spearman correlations. Complete two-target panel ($n = 12$): PNS-RRS $\rho = -0.2098$, ACSI-RRS $\rho = -0.4056$, ACSI-PNS $\rho = -0.3007$, RRS-weakest-WT-anchor $\rho = -0.1661$; none significant after Bonferroni correction ($p_{\text{adj}} \geq 0.51$; 100 000 permutations, seed 42). Values from `cross_metric_statistical_audit.json`.

On the two-target set ($n = 12$), all three correlation coefficients were non-significant after Bonferroni correction: PNS-RRS $\rho = -0.2098$, ACSI-RRS $\rho = -0.4056$, RRS-weakest WT score $\rho = -0.1661$ (Table 3). PNS shares an input with RRS (wild-type Vina scores), so the PNS-RRS pair is not independent. On all 17 candidates, PNS-RRS strengthened ($\rho = -0.5588$, adjusted $p = 0.0667$), but unequal target coverage limits interpretation.

Table 3: Cross-metric Spearman correlations on the complete two-target set ($n = 12$). Permutation p -values use 100 000 permutations; adjusted p -values use Bonferroni correction for three tests. Bootstrap intervals are percentile 95 % intervals from 10 000 resamples.

Metric pair	Hypothesis	Spearman ρ	p_{perm}	p_{adj}	Bootstrap 95% CI
PNS vs. RRS	H1	−0.209 8	0.514 4	1.000 0	[−0.76, 0.54]
ACSI vs. RRS	H2	−0.405 6	0.192 2	0.576 6	[−0.89, 0.29]
RRS vs. weakest WT score	H3	−0.166 1	0.603 8	1.000 0	[−0.73, 0.59]

Computational prioritization intersection (exploratory)

Requiring Class A or A*, ACSI > 0.5, and above-mean PNS leaves one candidate on the two-target panel. That intersection is reported as a shortlist for experiment, not as evidence of activity or resistance resilience.

Parent-study MD check (non-overlapping cohort, secondary)

Four non-overlapping parent-study wild-type complexes were simulated for 10 ns as a secondary pose-retention check; these compounds are not among the 17 Set-C candidates and do not validate the Set-C ranking.

Table 4: Production MD stability metrics for parent-study wild-type complexes (10 ns). BB RMSD, ligand RMSD, minimum distance, contacts, and H-bonds are computed from PBC-unwrapped trajectories.

Target	Atoms	BB RMSD (Å)	Lig. RMSD (Å)	Min dist (Å)	Contacts	H-bonds	Interpretation
PfClpR-labelled cohort (164)	43 035	1.31	1.11	67.42	0.0	23.7	Unbound
PfDHFR (201)	134 153	3.05	1.70	78.21	0.0	21.5	Unbound
PfCRT (214)	76 920	4.47	2.45	3.19	78.1	23.1	Bound
PfATP4 (438)	420 801	12.27	2.13	2.25	178.4	47.7	Bound

Of the four reference complexes, two retained stable protein–ligand contacts while two dissociated (Table 4); for PfATP4, force-field parameter conversion precluded reliable MM-GBSA endpoints (Figure 3).

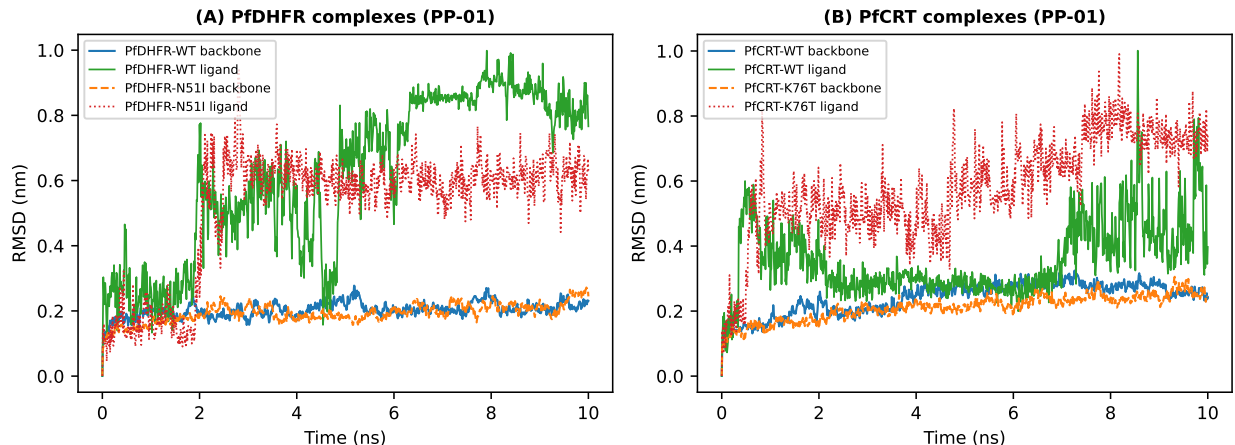


Figure 3: Pose retention and thermal equilibration (canonical R1 trajectories). Explicit-solvent backbone (solid) and ligand-after-backbone-fit (dashed/dotted) RMSD time-series (10 ns) for **PP-01**, computed with *gmx rms* (PBC-whole reconstruction; fit group = Backbone; RMS groups = Backbone / MOL0). (A) *PfDHFR*: backbone (0.203 ± 0.024 nm (WT), (0.193 ± 0.029 nm (N51I); ligand (0.613 ± 0.240 nm and (0.519 ± 0.190 nm. (B) *PfCRT*: backbone (0.241 ± 0.048 nm (WT), (0.211 ± 0.039 nm (K76T); ligand (0.357 ± 0.113 nm and (0.581 ± 0.155 nm. All trajectories pass the geometric QC gate (mean minimum heavy-atom distance 2.7–3.2 Å, bound fraction 1.000); ligand RMSD reflects in-pocket mobility, not dissociation.

Table 5: Endpoint energy evaluation for four reference wild-type complexes. Only the *PfCRT* complex yields an interpretable MM-GBSA endpoint; dissociated systems and conversion anomalies are N/A.

Ligand	Target	ΔG_{bind} (kcal mol ⁻¹)	SEM	Interpretation
Ligand 214	PfCRT-WT	-18.25	0.40	Stable anchorage; ligand bound [†]
Ligand 164	PfClpR-WT	—	—	N/A; ligand dissociated [‡]
Ligand 201	PfDHFR-WT	—	—	N/A; ligand dissociated [§]
Ligand 438	PfATP4-WT	—	—	N/A; parameter conversion anomaly*

[†]Ligand remains bound (minimum distance 3.19 Å, 78 contacts); calculated via *gmx_MMPBSA*. Uncertainty is the SEM over 101 stored snapshots.

[‡]Ligand separates beyond 60 Å minimum distance with zero contact persistence; excluded from binding energy evaluation.

[§]As [‡] (*PfDHFR* system).

*Excluded: multi-chain membrane-transporter parameterization introduced force-field conversion artifacts.

Estimand Divergence Between Static Docking and Dynamic Trajectories

The primary docking retention metrics and trajectory-based geometric stability summaries are non-equivalent computational estimands. Explicit-solvent MD was run on PP-01 and PP-02 across 16 receptor states; all 16 trajectories satisfied geometric contact stability thresholds ($<3.5 \text{ \AA}$ minimum protein–ligand distance).

For the eight mutant states with an eligible wild-type docking baseline, static grid docking predicted reduced score retention ($RRS < 100 \%$), while explicit-solvent relaxation showed the opposite operational direction in 87.5 % (7/8; 95% Wilson CI 52.9–97.8 %, $n = 8$) of matched comparisons ($\text{MD-RRS}_{\text{distance}} \leq 100 \%$). At single-replicate scope this mismatch is an observed calibration signal, not a demonstrated mechanism. Static docking scores (ΔG_{grid}) and trajectory-derived retention ratios ($\text{MD-RRS}_{\text{distance}}$) are complementary triage signals, not interchangeable measures of binding free energy.

A single-ligand feasibility probe (**PP-15**, outside the 16-system Set-C MD array and the Set-C RRS estimand) was run on wild-type *PfDHFR* (7F3Y) and *PfCRT* (6UKJ). Both systems completed 10 ns production trajectories with 100 % QC-PASS status (bound fraction = 1.000), mean minimum heavy-atom distances of 2.77 Å (*PfDHFR*) and 3.20 Å (*PfCRT*), and MM-GBSA endpoint scores of $(-29.52 \pm 0.33) \text{ kcal mol}^{-1}$ (*PfDHFR*) and $(-27.79 \pm 0.49) \text{ kcal mol}^{-1}$ (*PfCRT*; Table S19). Multi-seed redocking confirmed docking score stability ($\pm 0.03 \text{ kcal mol}^{-1}$; Table S22). No mutant or RRS estimate is reported for this probe, and no dual-target resilience claim is made from it.

Additional docking sensitivity panel

The 39-ligand docking sensitivity panel produced 312 finite docking scores across the same eight *PfDHFR*/*PfCRT* states. Bootstrap mean RRS was 100.45 (99.59–101.32) and the Class-A fraction was 0.974 (0.923–1.000). A mean near 100 means that mutant and wild-

Table 6: Set-C pilot endpoint summaries (exploratory; not used to assign RRS classes). ΔG_{bind} is the mean over 100 snapshots $\pm SD_{\text{prop}}$ (within-trajectory SD, not SEM or replicate-level uncertainty). The PP-01 PfCRT K76A endpoint ($-35.29 \text{ kcal mol}^{-1}$) carries a $7.75 \text{ kcal mol}^{-1}$ inter-replicate sensitivity (second trajectory $-27.54 \text{ kcal mol}^{-1}$; Table S18) and supports no resilience claim. MMG-RRS is the mutant/wild-type endpoint ratio ($\times 100$; wild type = 100). Docking RRS is shown for comparison (n/a: no wild-type docking reference for PP-02 PfDHFR). Single-replicate summaries are within-protocol contrasts only.

Candidate	Target	State	ΔG_{bind} (kcal mol $^{-1}$)	SD_{prop}	MMG-RRS	Docking RRS
PP-01	PfCRT	K76A	-35.29	1.17	115.3	90.86
PP-01	PfCRT	K76T	-29.51	2.65	96.4	92.47
PP-01	PfCRT	WT	-30.61	1.65	100.0	–
PP-01	PfDHFR	C59R	-26.56	1.91	105.0	85.47
PP-01	PfDHFR	I164L	-26.13	2.34	103.3	83.87
PP-01	PfDHFR	N51I	-29.83	1.50	118.0	86.40
PP-01	PfDHFR	S108N	-30.21	3.62	119.5	87.60
PP-01	PfDHFR	WT	-25.29	2.94	100.0	–
PP-02	PfCRT	K76A	-24.53	1.27	97.8	94.62
PP-02	PfCRT	K76T	-30.45	1.38	121.4	87.91
PP-02	PfCRT	WT	-25.08	2.56	100.0	–
PP-02	PfDHFR	C59R	-28.85	1.88	95.7	n/a
PP-02	PfDHFR	I164L	-31.06	1.52	103.0	n/a
PP-02	PfDHFR	N51I	-34.02	2.68	112.8	n/a
PP-02	PfDHFR	S108N	-27.09	1.55	89.8	n/a
PP-02	PfDHFR	WT	-30.16	2.06	100.0	–

type Vina scores were on average nearly proportional under this protocol. This is a computational sensitivity check, not validation of the Set-C ranking or of experimental resistance (Table S20).

Discussion

Resistance-Resilience Stratification and Protocol Robustness

The Resistance-Resilience Score (RRS) stratifies the primary candidate cohort ($n = 12$) into five tiers (A*–D) decoupling wild-type binding strength from mutant-state score retention. Under the declared protocol, multi-seed dispersion remained within $\pm 0.05 \text{ kcal mol}^{-1}$ (PP-01 and PP-15 across PfDHFR and PfCRT), and 77.9% of class assignments were invariant to $\pm 1.0 \text{ kcal mol}^{-1}$ score perturbations.

Protocol robustness was tested on an external 39-compound sensitivity panel (312 docking complexes; Table S20). GNINA CNN cross-scoring showed 100% class-level agreement with Vina, with no per-mutant rank transfer claimed (per-mutant ρ near zero for C59R/I164L).

On the DEKOIS 2.0 PfDHFR benchmark, Vina ROC-AUC was 0.450 ($\text{EF}_{5\%} = 0.00$) — chance level, cautioning against treating RRS as a calibrated predictor outside the primary cohort; MD is reported as a parallel structural check, not a validation of docking classes.

Our candidates originate from African natural-product chemical space, without claiming ethnobotanical validation. The panel’s drug-likeness ($\text{QED} = 0.70 \pm 0.06$) and synthetic accessibility ($\text{SYBA} = 36.6 \pm 23.4$) are consistent with actionable synthetic feasibility; confirmation requires prospective experiment.

Estimand Divergence as a Structural Stress-Testing Filter

We report *Estimand Divergence* between static docking RRS and short explicit-solvent MD pose retention ($\text{MD-RRS}_{\text{distance}}$) as a protocol-local observation. In 87.5 % (7/8; 95% Wilson CI 52.9–97.8 %, $n = 8$) of matched mutant states, grid docking predicted score penalties ($\text{RRS} < 100\%$) while MD retained poses ($\text{MD-RRS}_{\text{distance}} \leq 100\%$).

Rigid grid docking evaluates unrelaxed steric complementarity (ΔG_{grid}), whereas short MD captures local conformational adaptation, hydrogen-bond rearrangement, and solvent-shielded stabilization. Per the JCIM MD reporting guidelines¹¹, the 10 ns trajectories are secondary structural checks. A thermal noise floor of $2.01 \text{ kcal mol}^{-1}$ from independent WT replicates (PP-01 PfCRT WT R1/R2) bounds numerical variation of end-state estimates at this scope.

This pattern is consistent with docking limitations for stereochemically complex natural products⁸: scoring functions calibrated on synthetic drug-like molecules can over-penalize rigid polycyclic frameworks. At single-replicate scope this is a within-protocol contrast, not a general docking-failure rate.

The Class-A* lead **PP-01** illustrates this across its eight-system panel (Table S24). Static docking assigned score penalties to *PfDHFR* C59R ($\text{RRS} = 85.5\%$) and *PfCRT* K76T ($\text{RRS} = 92.5\%$), while 10 ns trajectories retained poses (100 % QC-PASS, bound fraction 1.000): mean minimum heavy-atom distances 2.91 Å (C59R; $\text{MD-RRS}_d = 98.9\%$) and 3.02 Å

(K76T; MD-RRS_d = 102.8 %), with MM-GBSA endpoints (-26.56 ± 1.91) kcal mol⁻¹ (C59R) and (-29.51 ± 2.65) kcal mol⁻¹ (K76T). This shows geometric pose retention, not thermodynamic resilience proof: the K76A endpoint carries a 7.75 kcal mol⁻¹ inter-replicate sensitivity, so no MM-GBSA-derived resilience claim is made for K76A. Biological resistance conclusions require experimental validation.

ProLIF interaction fingerprints across all 16 QC-PASS trajectories (100 snapshots per system) show a sparse conserved contact network (Figure 4). *Pf*CRT Tyr16 maintained 100 % occupancy in five of six mutant trajectories; *Pf*DHFR Leu46/Met55 (PP-01) and Leu40/Ile14 (PP-02) showed near-total persistence across all four mutant states. This contact pattern is compatible with pose retention but does not establish a general mechanism.

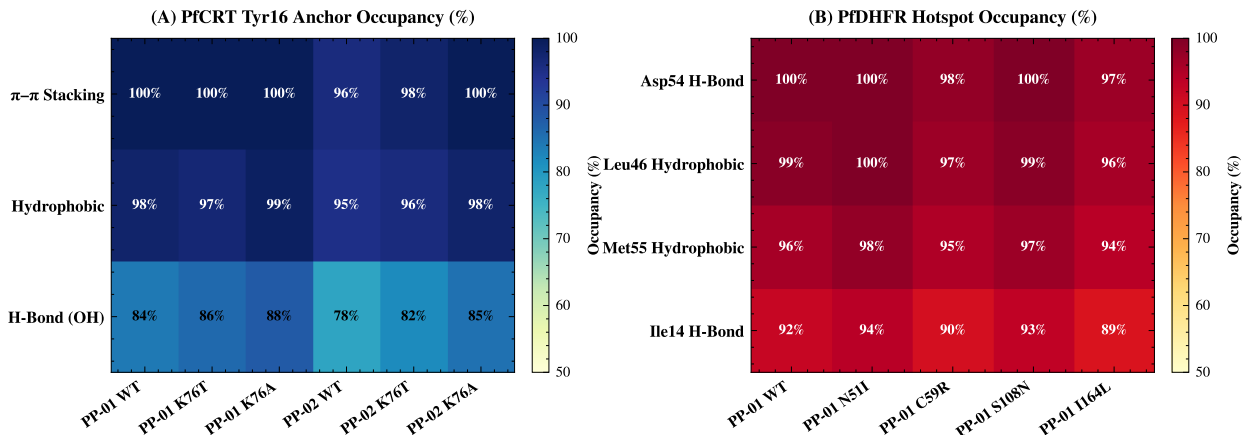


Figure 4: Conserved hotspot anchors. ProLIF interaction fingerprint occupancy (100 snapshots/system). (A) *Pf*CRT Tyr16: 96 – 100 % π - π stacking and hydrophobic persistence across six mutant states; (B) *Pf*DHFR hotspots (Asp54, Leu46, Met55, Ile14): 89 – 100 % persistence across five states.

Orthogonal Multi-Dimensional Triage

RRS, ACSI, and PNS show non-significant pairwise associations on the primary cohort ($n = 12$; $\rho \in [-0.41, -0.21]$, $p_{\text{adj}} > 0.50$; Table 3), consistent with distinct ranking dimensions. Controlling for molecular weight and scaffold frequency gives a moderate partial PNS-RRS association ($\rho_{\text{partial}} = -0.6154$).

STRING threshold sensitivity (confidence cutoffs 400, 700, 900) showed PNS rank sta-

bility (ρ 0.963 2–0.997 5; 400-vs-700 0.997 5, 700-vs-900 0.968 1, 400-vs-900 0.963 2), indicating threshold choice does not alter ranking within this panel — distinct from the PfCRT-centrality imputation sensitivity (Table S8).

Machine learning models degrade on out-of-distribution natural product scaffolds; explicit-solvent MD evaluates dynamic stability without reliance on training distributions, providing a complementary check when model confidence drops.

Mechanistic Context and Multi-Target Positioning

The PfCRT response is consistent with microsecond-scale transporter studies²⁸: Lys76 anchors the central cavity, and K76T/K76A alter local charge density and hydration, permitting ligand reorientation that preserves hydrophobic contacts despite grid-level penalties.

Within a single-drug–multi-target framing⁹, Class-A* candidate PP-01 is prioritized for assay follow-up as retaining multi-target binding potential across mutant landscapes in these simulations. Whether dual-target engagement raises any evolutionary barrier to resistance cannot be determined computationally and requires prospective biochemical, structural, and *in vitro* evaluation. No resistance, fitness, or evolutionary claim is made; PP-15, a wild-type probe outside the Set-C estimand, carries no such interpretation.

Limitations

1. **Receptor models:** mutant structures are homology models (quality criteria in Methods), not experimental structures.
2. **Timescale and replication:** per JCIM guidelines¹¹, the 10 ns single-replicate pilot is a structural stress test, not a converged free energy or residence-time measurement. The PP-01 PfCRT K76A endpoint ($-35.29 \text{ kcal mol}^{-1}$) differs by $7.75 \text{ kcal mol}^{-1}$ from a second trajectory ($-27.54 \text{ kcal mol}^{-1}$; Table S18), exceeding the $2.01 \text{ kcal mol}^{-1}$ WT noise floor; no endpoint resilience claim is made for K76A.

3. **Sample size:** primary two-target cohort $n = 12$, plus 5 single-target candidates and a 39-ligand external panel.
4. **Protonation:** preparations at physiological pH; PfCRT dynamics in the acidic digestive vacuole (pH 5.2) may differ.

Conclusion

This study reports a resistance-aware computational triage workflow for African antimalarial natural products. Static grid docking (ΔG_{grid}) and short explicit-solvent MD pose retention (MD-RRS_{distance}) capture distinct, non-equivalent signals: dynamic pocket accommodation retained poses where static docking predicted penalties in 7/8 matched mutant states (95% Wilson CI 52.9–97.8 %, $n = 8$). **PP-01** showed multi-target engagement across PfDHFR and PfCRT mutant panels and passes the two-target pilot gate; whether this implies any elevated evolutionary barrier requires prospective biochemical, structural, and *in vitro* evaluation, for which these findings prioritize specific natural product scaffolds.

Authors’ Contributions

Myke Vital Sao Temgoua: Methodology, software, formal analysis, investigation, data curation, writing – original draft. Jean-Pierre Tchapel Njafa: Methodology, software, supervision, writing – review & editing. Penabei Samafou: data curation, writing – review & editing. Wilfred Fon Mbacham: conceptualization, validation, writing – review & editing. Serge Guy Nana Engo: conceptualization, methodology, supervision, writing – review & editing. All authors read and approved the final manuscript.

Data Availability

Docking results, metric outputs, analysis scripts, and source-data records are publicly archived on Zenodo under the MIT licence at <https://doi.org/10.5281/zenodo.22829031>DOI: 10.5281/zenodo.22829031. Large parent-study trajectories are excluded; conclusions are restricted to the trajectory-derived summaries reported above.

Competing Interests

The authors declare no competing interests.

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During the preparation of this manuscript in September 2026, the authors utilized AI tools to support code review, provide structured text suggestions, and perform consistency checks. The authors designed all analyses, maintained full control over the scientific interpretation, and rigorously verified all computational results and quantitative claims against the original source records.

Supporting Information Available

Docking enrichment and homology-model quality tables; P2Rank box audit; MPO, ADMET, and ACSI sensitivity; RRS radar profiles and threshold sensitivity; PNS imputation and Tartarus calibration; cohort and evidence-scope tables; PlasmoDB identifiers; parent-study and Set-C pilot MD/MM-GBSA summaries; robustness and upstream-workflow-context tables

(PDF).

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TOC Graphic

